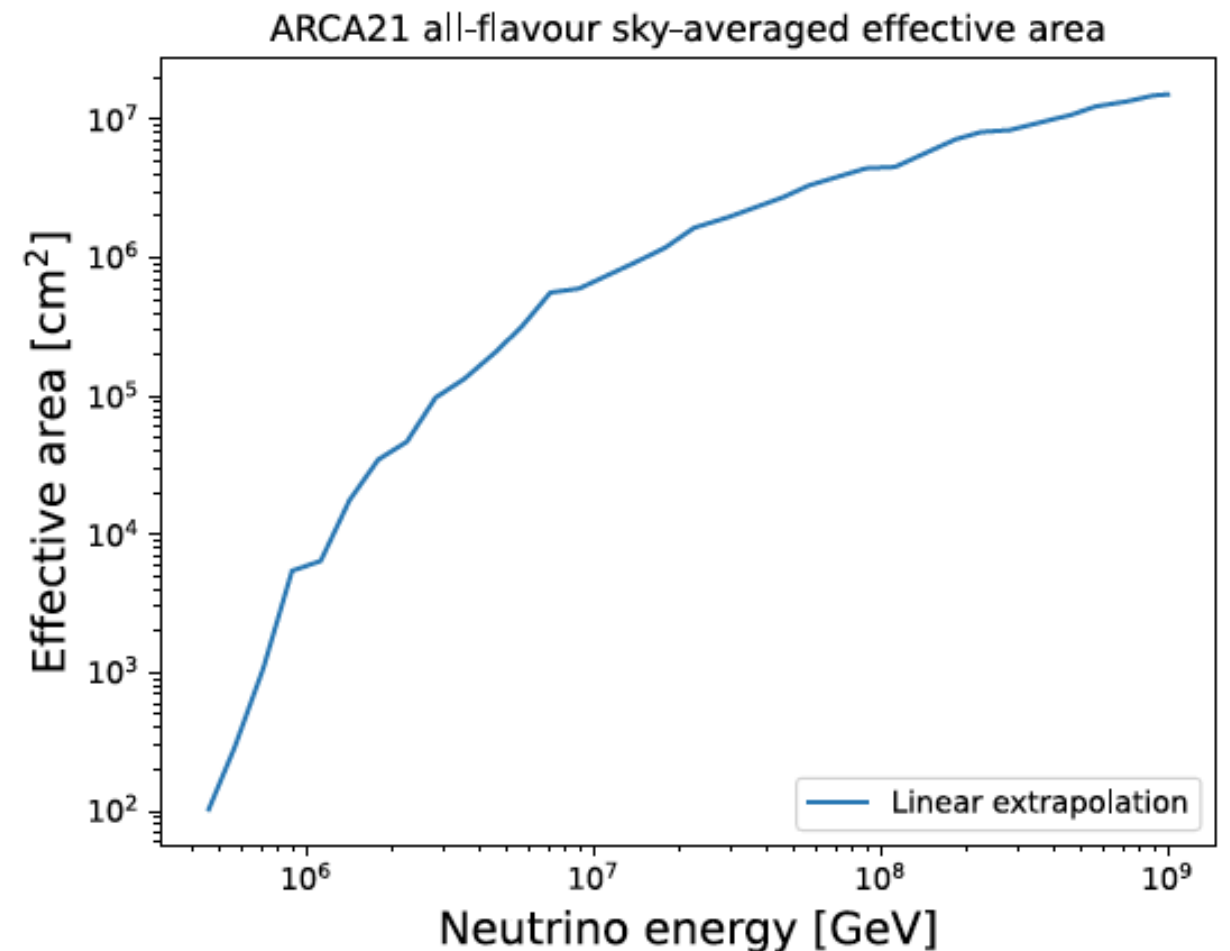


Analysis plan

- We are working on the unbinned with a real period of ARCA21.
- ARCA230 in 10 years ... -> we want to study supernovae for now -> use ARCA21.
- The effective area of ARCA21 ([LINK](#)) should to be the one used for the VHE by our group during the summer.
- We can use the EA because is independent if the source is transient or not.
- To get the expected number of events:



(convolution of the neutrino flux with the EA)* lifetime

→ *Murase neutrino flux*

Expected number of neutrinos

$$N_{\nu} = \int_{t_{min}}^{t_{max}} \int_{E_{min}}^{E_{max}} \phi(E_{\nu}, t) A_{eff}(E_{\nu}) dt dE_{\nu}$$

Integrated SN neutrino flux
over time and energy

Integrated effective area
over energy

Script: /sps/km3net/users/mbendahman/HE-CCSN/Effective_Area/aeff.py

Detector response: 202475_5_49_54_DETECTORRESPONSE_ARCA21_133_tmva_tracks_v10.ALL_zen_b40_pre_upm01_antinoise_bdt050.root
202475_5_53_29_DETECTORRESPONSE_ARCA21_133_tmva_tracks_v10.ALL_zen_b40_pre_upm01_antinoise_bdt095.root

1. Calculate the effective area at the declination of the source.
2. Calculate the number of neutrino events.